Oldham College

Networking Academy

LAB WORKSHEET

VLAN Practice Lab

A focused two-switch lab for VLANs and trunking. Guided objectives tick off automatically as you configure and test network segmentation.

The scenario

Two switches (SW1, SW2) are joined by a trunk. Four PCs are already addressed on 192.168.1.11–14 /24. Your job is to put them into VLANs and prove the segmentation works.

Objectives

1. Create **VLAN 10 (STAFF)** and **VLAN 20 (STUDENT)** on both switches.
2. Put PC1 & PC3 in VLAN 10, PC2 & PC4 in VLAN 20.
3. Configure the link between the switches as an **802.1Q trunk**.
4. Test: same-VLAN PCs should ping; different-VLAN PCs should not.

Configure VLANs (repeat on SW1 and SW2)

```
enable
configure terminal
 vlan 10
  name STAFF
 vlan 20
  name STUDENT
 exit
```

Assign access ports

```
 interface f0/1
  switchport mode access
  switchport access vlan 10
 interface f0/2
  switchport mode access
  switchport access vlan 20
 exit
```

Build the trunk

```
 interface g0/1
  switchport mode trunk
 end
show vlan brief
show interfaces trunk
```

Test segmentation

- From PC1 `ping` PC3 — should **succeed** (both VLAN 10).
- From PC1 `ping` PC2 — should **fail** (different VLANs, no router).

Why it fails: VLANs are separate broadcast domains. To let them talk you would need inter-VLAN routing (a router-on-a-stick or a Layer-3 switch) — a good extension task.